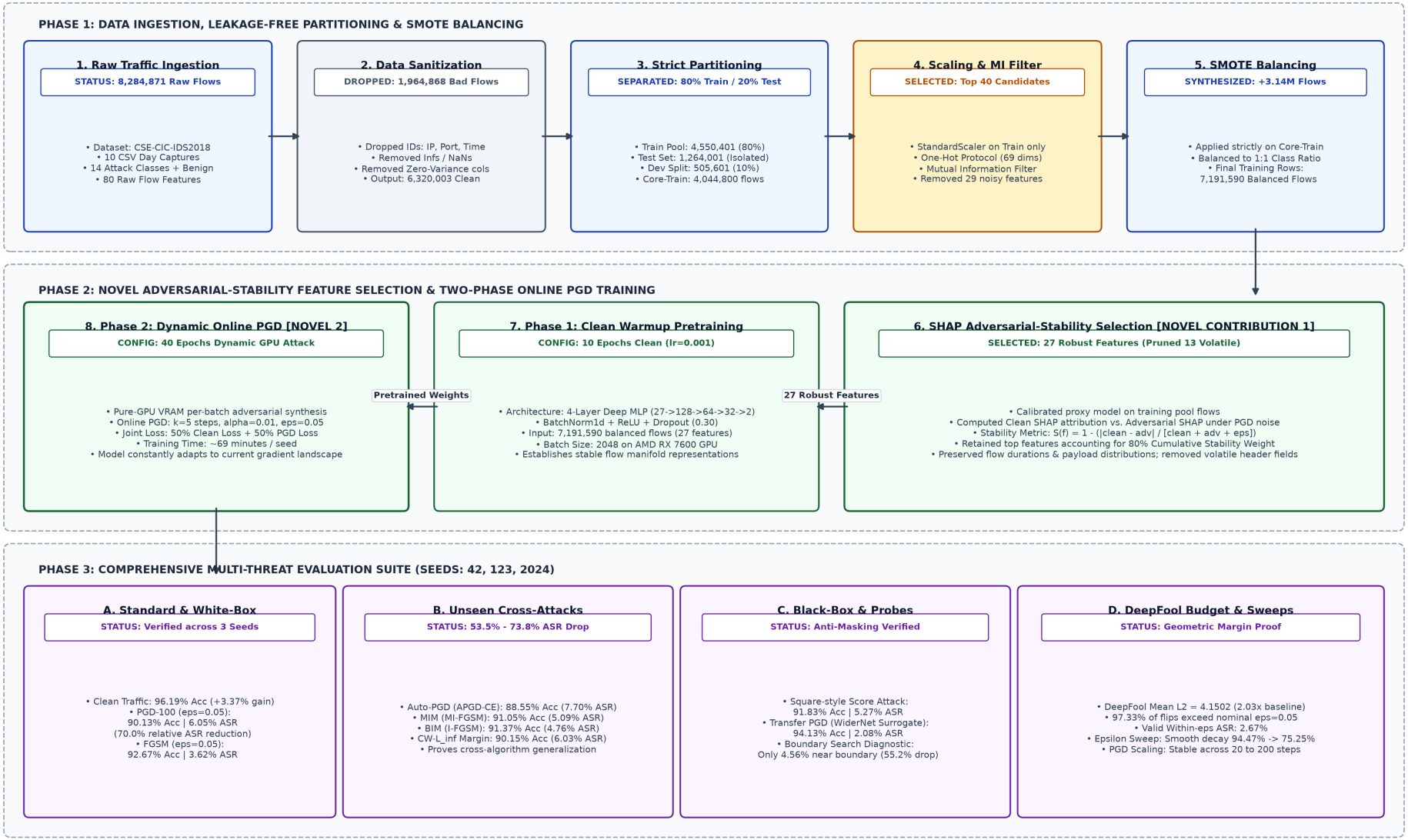
Proposed Methodology Architecture: Adversarial Robust Cloud NIDS (V9)

Benchmark: CSE-CIC-IDS2018 (8.28M Raw / 6.32M Clean Flows) | **Evaluation:** 3-Seed Replication (42, 123, 2024) | **Hardware:** AMD Radeon RX 7600 (ROCm / PyTorch)

Proposed Methodology Flowchart: Adversarial Robust Cloud NIDS (V9 Defense)

End-to-End Pipeline: Data Ingestion, Strict Partitioning, SHAP Stability Feature Selection & Online PGD Hardening



Methodology Pipeline Status & Configuration Breakdown (V9 Defense)

Complete quantitative ledger of data transitions, feature dimensions, training parameters, and multi-threat evaluations:

1. Data Pipeline, Partitioning & Feature Selection Ledger

Pipeline Step	Input Status	Operation & Method	Dropped / Added	Output Status	Configuration / Rationale
1. Raw Ingestion	8,284,871 flows 80 raw columns	CSE-CIC-IDS2018 AWS benchmark ingestion across 10 CSV days	None (0 dropped)	**8,284,871 flows**	14 intrusion attack categories + authentic benign traffic
2. Sanitization	8,284,871 flows	Dropped IDs (`Timestamp`, `IP`, `DstPort`), `+inf`/`-inf`/`NaNs`, zero-variance	**-1,964,868** bad flows	**6,320,003 Clean**	Prevents shortcut memorization of timestamps and destination ports
3. Partitioning	6,320,003 flows	Stratified 80/20 train/test split; Train divided into 90% Core / 10% Dev	Zero leakage (Isolated)	Train: **4,550,401** Test: **1,264,001**	Test partition is sealed immediately; zero leakage to preprocessing
4. MI Candidate Selection	69 one-hot dims	StandardScaler fitted on Train; Mutual Information (MI) scoring	**-29** noisy features	**40 Candidate Dims**	Eliminates irrelevant noise dimensions prior to oversampling
5. SMOTE Balancing	4,044,800 Core-Train	Synthetic Minority Over-sampling applied strictly on Core-Train	**+3,146,790** synthetic	**7,191,590 Balanced**	Equalized 1:1 ratio across all 14 minority classes on train pool only
6. SHAP Stability Selection	40 Candidate Dims	Adversarial Stability Metric: S(f) = 1 - (|clean - adv| / denom)	**-13** fragile features	**27 Robust Dims**	Retains 80% cumulative stability weight; prunes fragile header fields

2. Deep Learning Architecture, Two-Phase Training & Multi-Threat Results

Pipeline Component	Detailed Specifications & Parameters	Observed Performance & Empirical Validation
Model Architecture	4-Layer Deep MLP: Input(27) → 128 → 64 → 32 → Output(2 Logits) BatchNorm1d + ReLU Activation + Dropout(0.30) after each hidden layer	Optimized for fast per-batch gradient ascent on AMD Radeon RX 7600 GPU (8GB VRAM)
Phase 1: Warmup Training	10 Epochs on clean balanced data (7,191,590 rows) | Batch Size: 2048 | Adam (lr=0.001)	Establishes stable foundational flow manifold and baseline decision boundary
Phase 2: Dynamic Online PGD	40 Epochs Pure-GPU real-time adversarial synthesis | k=5 steps, alpha=0.01, eps=0.05 Joint Loss Objective: 50% Clean Cross-Entropy + 50% Adversarial Cross-Entropy	Training Time: ~69 min / seed | Adapts to instantaneous gradient vulnerabilities
Standard & White-Box Defense	Evaluated on held-out test partition (1,264,001 flows / 2k slice) across Seeds [42, 123, 2024]	**Clean Acc: 96.19% ± 0.03% (+3.37% gain) | PGD-100 Acc: 90.13% (6.05% ASR) | FGSM Acc: 92.67%**
Unseen Cross-Attack Defense	Evaluated against Auto-PGD (APGD-CE), MIM (mu=1.0), BIM (clean start), CW-L_inf margin	**APGD Acc: 88.55% (7.70% ASR) | MIM Acc: 91.05% | BIM Acc: 91.37% | CW Acc: 90.15%**
Black-Box & Margin Proof	Square-style score search, Transfer PGD (WiderNet surrogate), Unconstrained DeepFool	**Transfer Acc: 94.13% (2.08% ASR) | DeepFool Mean L2 = 4.1502 (97.33% flips exceed eps=0.05)**